DUAL - BAND SLOTTED MICROSTRIP ANTENNA FOR SMART FARMING COMMUNICATION SENSOR

ABSTRACT

The design and analysis of a compact dual-band slotted microstrip antenna for smart agricultural applications. The increasing demand for efficient wireless communication in IoT-based farming systems necessitates reliable antenna solutions. The proposed antenna is designed to operate at two frequency bands, namely 2.4 GHz and 4.3 GHz. These frequency bands are suitable for wireless sensor networks used in agricultural monitoring systems. The antenna structure consists of a rectangular microstrip patch with strategically introduced slots. These slots create multiple current paths, enabling dual-band resonance. The antenna is designed and simulated using ANSYS HFSS. Simulation results show a return loss of −29.43 dB at 2.4 GHz. At 4.29 GHz, the antenna achieves a return loss of −35.77 dB. The Voltage Standing Wave Ratio (VSWR) is observed to be close to unity at both frequencies. The antenna also demonstrates impedance values close to 50 ohms, indicating good matching. A stable quasi-omnidirectional radiation pattern is achieved, ensuring wide coverage. The design is compact, cost-effective, and suitable for practical implementation. It supports reliable communication for smart irrigation and crop monitoring systems. Thus, the proposed antenna proves to be an efficient solution for smart agriculture applications.

Abstract (Key words)
Dual-Band Antenna, Microstrip Patch Antenna, Slotted Antenna, Smart Agriculture, Internet of Things (IoT), Wireless Sensor Networks (WSN), Return Loss, VSWR, HFSS, Impedance Matching

SDG Goal		Remarks
SDG 2	This project supports sustainable agriculture by enabling smart monitoring systems that improve crop yield and resource efficiency.	Enhances smart agricultural monitoring and enables efficient wireless communication for improved crop management.

Project Component	Relevant IEEE Standards
Dual-band antenna design	IEEE 145, IEEE 149
Microstrip antenna performance	IEEE 149
Wireless communication (IoT sensors)	IEEE 802.11, IEEE 802.15.4
RF performance & measurements	IEEE 1720, IEEE 473
Antenna testing & radiation analysis	IEEE 149
IoT system architecture	IEEE 2413
System reliability & performance	IEEE 1633
Electromagnetic compatibility (EMC)	IEEE 299

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LIST OF ABBREVIATIONS

ACRONYM		ABBREVIATIONS
BW	-	Bandwidth
DGS	-	Defected Ground Structure
EM	-	Electromagnetic Spectrum
HFSS	-	High Frequency Structure Simulator
IoT	-	Internet of Things
ISM	-	Industrail, Scientific and Medical
PCB	-	Printed Circuit Board
RF	-	Radio Frequency
VSWR	-	Voltage Standing Wave Ratio
WSN	-	Wireless Sensor Network

CHAPTER 1

INTRODUCTION

1.1 SMART AGRICULTURAL SYSTEMS

In recent years, agriculture has evolved significantly with the integration of advanced technologies such as the Internet of Things (IoT), wireless sensor networks (WSNs), and automated monitoring systems. Smart agricultural systems aim to improve productivity, reduce resource wastage, and enable real-time decision-making through continuous monitoring of environmental parameters. Sensors deployed across agricultural fields measure factors such as soil moisture, temperature, humidity, and nutrient levels, which are then transmitted to centralized systems for analysis. Reliable wireless communication is a key requirement in such systems, as data must be transmitted efficiently over large areas.

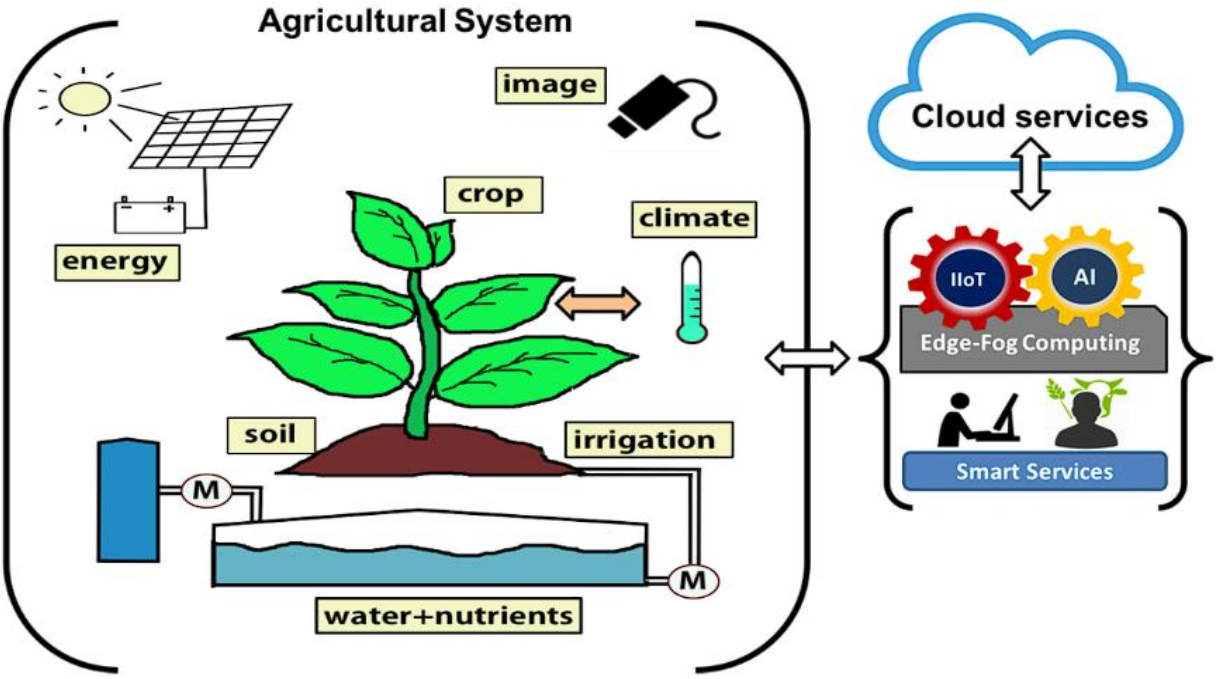

Figure 1.1 Smart Agriculture with Data Transmission

Key Features of Smart Agriculture:

- Real-time monitoring of field conditions
- Automated irrigation systems
- Data-driven decision making
- Integration with IoT platforms

1.2 ELECTROMAGNETIC (EM) SPECTRUM

The electromagnetic spectrum represents the range of all electromagnetic waves arranged according to their frequency or wavelength. It includes various regions such as radio waves, microwaves, infrared, visible light, ultraviolet, X-rays, and gamma rays. Among these, the radio frequency (RF) and microwave regions are widely used for wireless communication applications. The proposed antenna operates in the microwave region of the electromagnetic spectrum, specifically at frequencies of 2.4 GHz and 4.3 GHz.

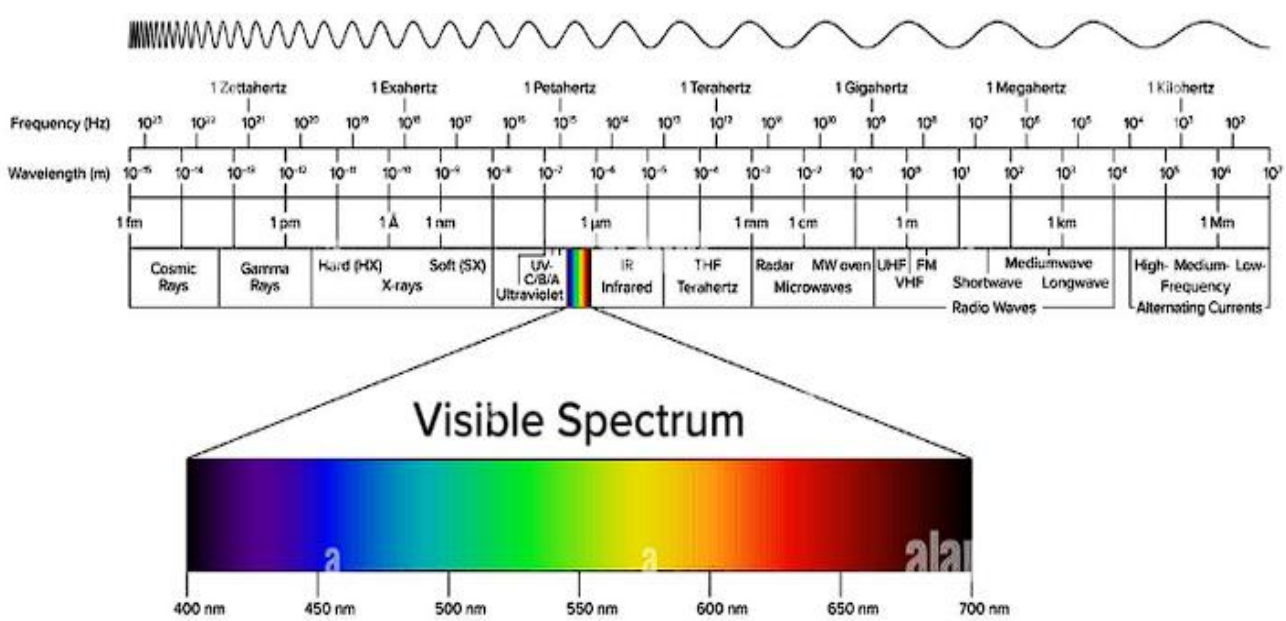

Figure 1.2 Electromagnetic Spectrum

The figure 1.2 shows the electromagnetic spectrum arranged based on frequency and wavelength. The proposed antenna operates in the microwave region at 2.4 GHz and 4.3 GHz, which fall within the ISM band. These frequencies are suitable for wireless communication in smart agricultural applications, enabling reliable data transmission between sensor nodes.

Important Frequency Ranges:

- RF Range: 3 kHz – 300 GHz
- Microwave Range: 1 GHz – 30 GHz

Relevance to the Project:

- Operates in GHz range for the High-speed communication
- Suitable for IoT-based systems
- Provides reliable signal propagation

1.3 RADIO FREQUENCY (RF) AND RF DEVICES

Radio Frequency (RF) refers to electromagnetic waves used for wireless communication. RF technology enables the transmission of signals through free space without the need for physical connections. Various RF devices are used to generate, transmit, receive, and process these signals.

In the proposed system the antenna acts as a key RF component that ensures efficient signal transmission and reception. The performance of the antenna directly affects the efficiency of the entire RF communication system.

Common RF Devices:

- Transmitters
- Receivers
- Antennas
- Amplifiers
- Filters

1.4 ANTENNA

An antenna is a device that converts electrical signals into electromagnetic waves during transmission and converts electromagnetic waves back into electrical signals during reception. It is one of the most important components in any wireless communication system.

Antennas are designed to radiate energy efficiently and ensure reliable communication over a specific frequency range. Based on their structure and application, antennas can be classified into various types such as wire antennas, microstrip antennas, and reflector antennas. In this project, a slotted microstrip patch antenna is used due to its compact size, ease of fabrication, and suitability for IoT applications.

Functions of an Antenna:

- Transmission of signals
- Reception of signals
- Radiation of electromagnetic energy

Advantages of Microstrip Antennas:

- Low profile
- Lightweight
- Cost-effective
- Easy integration with circuits

1.5 ANTENNA PARAMETERS

The performance of the proposed antenna is evaluated using several important parameters. These parameters help in determining the efficiency, reliability, and suitability of the antenna for smart agricultural applications.

1.5.1 RETURN LOSS (S11)

Return loss indicates the amount of power reflected back from the antenna due to impedance mismatch.

$$S_{11} = 20\log\left(\frac{V_{reflected}}{V_{incident}}\right) \quad (1)$$

- Good antenna performance requires S11 < −10 dB

1.5.2 VOLTAGE STANDING WAVE RATIO (VSWR)

VSWR measures the efficiency of power transmission between the antenna and transmission line.

$$VSWR = \frac{1+|\Gamma|}{1-|\Gamma|} \quad (2)$$

- Ideal value = 1
- Acceptable ≤ 2

1.5.3 IMPEDANCE (Z)

Impedance represents the opposition offered to current flow.

$$Z = R + jX \quad (3)$$

- Ideal impedance = **50 Ω**

1.5.4 BANDWIDTH (BW)

Bandwidth is the range of frequencies over which the antenna operates effectively.

$$BW = f_{high} - f_{low} \quad (4)$$

1.5.5 GAIN

Gain indicates how efficiently the antenna radiates energy in a specific direction.

$$G = \eta \times D \tag{5}$$

Where:

- η = Efficiency
- D = Directivity

1.5.6 RADIATION PATTERN

The radiation pattern of an antenna is a graphical representation that shows how the antenna radiates electromagnetic energy in different directions. It indicates the strength of the signal radiated or received by the antenna as a function of direction. Radiation patterns are usually plotted in either two-dimensional (2D) or three-dimensional (3D) forms. In most cases, the pattern is represented in polar coordinates, where the radius indicates the power level and the angle represents the direction.

CHAPTER 2

MICROSTRIP ANTENNA

2.1 INTRODUCTION TO MICROSTRIP ANTENNA

A microstrip antenna is a type of low-profile antenna that consists of a radiating patch mounted on a dielectric substrate with a ground plane on the opposite side. These antennas are widely used due to their compact size, lightweight structure, and ease of fabrication. They operate based on the principle of electromagnetic radiation generated by the oscillating currents on the patch surface.

Microstrip antennas figure 2.1 are highly suitable for wireless communication systems, including satellite communication, mobile devices, and IoT applications. Their planar structure makes them easy to integrate with electronic circuits. However, conventional microstrip antennas suffer from limitations such as narrow bandwidth and low gain.

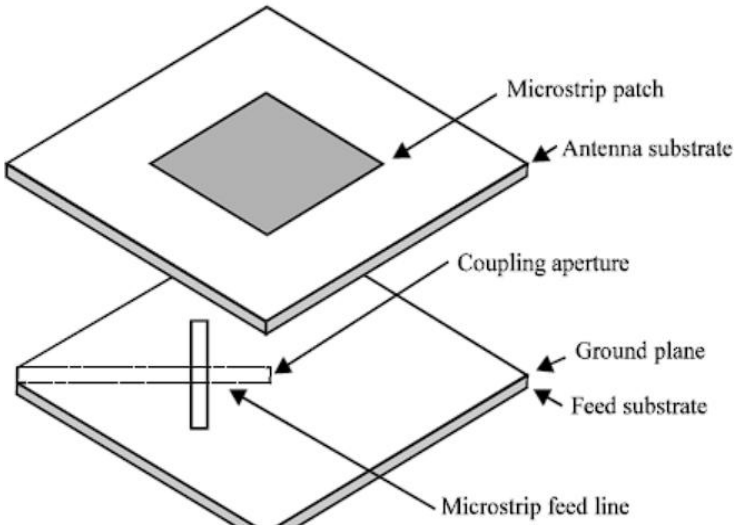

Figure 2.1 Microstrip Antenna

2.2 BASIC STRUCTURE OF MICROSTRIP ANTENNA

The basic structure of a microstrip antenna consists of three main components: the radiating patch, dielectric substrate, and ground plane. The patch is usually made of a conducting material such as copper and is responsible for radiating electromagnetic waves.

The substrate is a dielectric material placed between the patch and ground plane, characterized by its dielectric constant (εr) and thickness (h). The ground plane reflects the radiated waves and helps in forming the radiation pattern. The patch can be of various shapes such as rectangular, circular, triangular, or elliptical depending on the application.

The feed mechanism, such as microstrip line or coaxial probe, is used to excite the antenna. The dimensions of the patch determine the resonant frequency of operation. Proper design of these components ensures efficient radiation and impedance matching. The simplicity of the structure allows easy fabrication using printed circuit board (PCB) technology. This makes microstrip antennas highly suitable for compact and portable devices. The structure also supports modifications such as slotting to enhance performance.

2.3 SLOTTED MICROSTRIP ANTENNA

A slotted microstrip antenna is a modified version of the conventional microstrip antenna where slots are introduced on the patch or ground plane. These slots help in altering the current distribution on the patch surface, which in turn affects the resonant frequency and bandwidth. By introducing slots of different shapes and sizes, multiple current paths are created, enabling multi-band or dual-band operation.

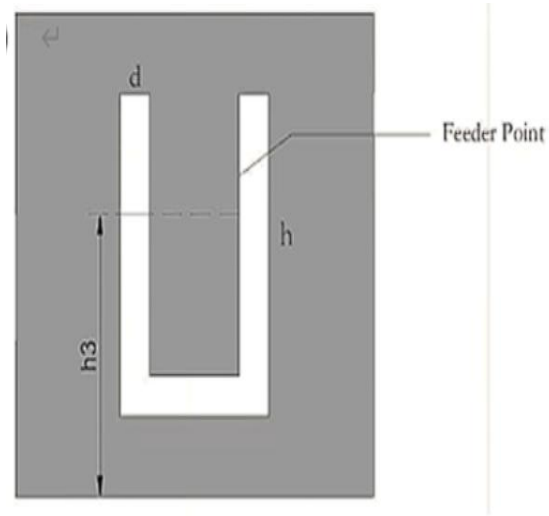

Figure 2.2 Slotted microstrip antenna

Slotted antennas are widely used to improve bandwidth, impedance matching, and gain. The slot design plays a crucial role in determining the antenna performance. Common slot shapes include rectangular, circular, U-shaped, E-shaped, and T-shaped slots. These slots increase the effective length of the current path without increasing the overall size of the antenna.

This results figure 2.2 in a compact design with enhanced functionality. Slotted microstrip antennas are highly suitable for modern wireless systems where multiple frequency bands are required. They are widely used in applications such as Wi-Fi, Bluetooth, and IoT-based systems. Overall, slotting is an effective technique for improving antenna performance.

2.4 TYPES OF MICROSTRIP ANTENNAS

Microstrip antennas can be classified into various types based on their shape, feeding technique, and application. Common types include rectangular patch antennas, circular patch antennas, triangular patch antennas, and elliptical patch antennas. Each type has its own advantages and is selected based on the required performance.

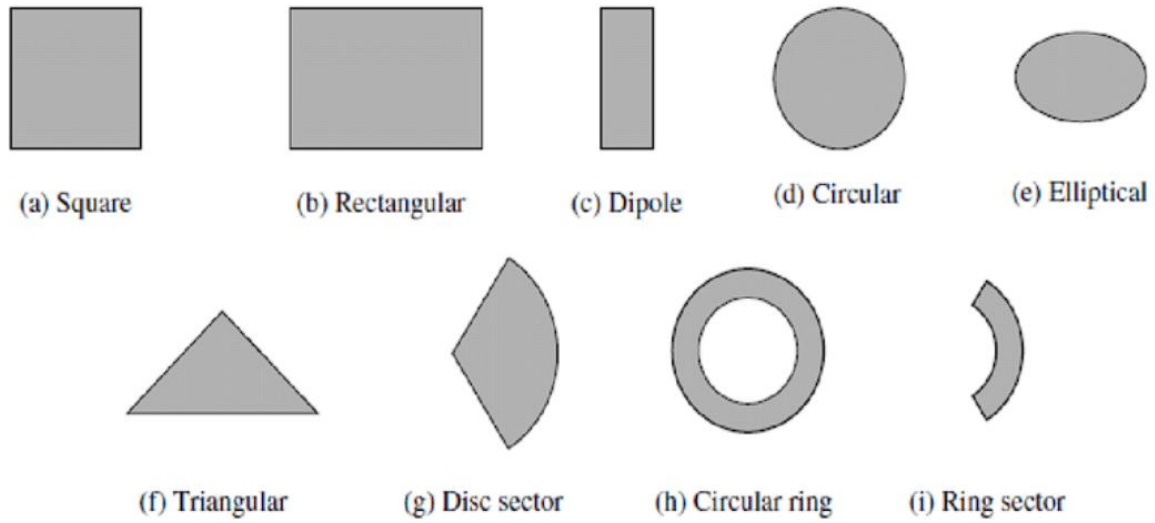

Figure 2.3 Types of slots

Rectangular patch antennas are the most widely used due to their simple design and ease of analysis. Circular patch antennas provide better polarization characteristics, while triangular antennas offer compact size. Another classification is based on feeding methods, such as microstrip line feed, coaxial probe feed, aperture coupling, and proximity coupling.

Additionally, antennas can be categorized based on their functionality figure 2.3, such as single-band, dual-band, and multi-band antennas. Slotted antennas fall under the category of modified microstrip antennas. Advanced designs also include reconfigurable antennas and fractal antennas. The selection of antenna type depends on factors such as frequency, bandwidth, size, and application requirements. This diversity allows microstrip antennas to be used in a wide range of communication systems.

Common slot shapes include rectangular, circular, U-shaped, E-shaped, and T-shaped slots. These slots increase the effective length of the current path without increasing the overall size of the antenna

2.5 ADVANTAGES OF SLOTTED MICROSTRIP ANTENNA

Slotted microstrip antennas offer several advantages over conventional antennas, making them highly suitable for modern communication systems. One of the main advantages is their ability to achieve multi-band or dual-band operation within a compact size.

The introduction of slots increases the effective current path, which helps in reducing the size of the antenna. These antennas also provide improved bandwidth compared to standard microstrip antennas. Another advantage is better impedance matching, which results in reduced reflection losses. Slotted antennas are lightweight, low-cost, and easy to fabricate using standard PCB techniques. They can be easily integrated with other electronic components, making them ideal for IoT applications.

The flexibility in slot design allows customization based on specific requirements. Additionally, these antennas offer good radiation efficiency and stable performance. They are widely used in wireless communication systems such as Wi-Fi, Bluetooth, and sensor networks. Overall, slotted microstrip antennas provide a balance between performance and compactness.

2.6 LIMITATIONS OF MICROSTRIP ANTENNAS

- Despite their advantages, microstrip antennas also have certain limitations that must be considered during design.
- One of the major drawbacks is their narrow bandwidth, which limits their performance in wideband applications.
- They also exhibit relatively low gain compared to other types of antennas. Surface wave losses in the substrate can reduce radiation efficiency.

- The dielectric substrate material plays a crucial role, and improper selection can degrade performance.
- The presence of spurious radiation can also affect the overall efficiency.
- The power handling capability, which is lower compared to conventional antennas. These issues can be mitigated using techniques such as slotting, stacking, and the use of advanced materials.

2.7 APPLICATIONS OF SLOTTED MICROSTRIP ANTENNA

Slotted microstrip antennas are widely used in various applications due to their compact size and multi-band capability. One of the major applications is in wireless communication systems such as Wi-Fi, Bluetooth, and mobile networks. They are also used in satellite communication and radar systems. In recent years, these antennas have gained significant importance in IoT-based applications, including smart agriculture.

In agricultural systems, they are used for monitoring environmental conditions such as temperature, humidity, and soil moisture. They enable reliable communication between sensor nodes and control systems. Slotted antennas are also used in medical devices, automotive systems, and wearable technology. Their ability to operate at multiple frequencies makes them suitable for modern communication systems.

Additionally, they are used in aerospace applications. The versatility and efficiency of slotted microstrip antennas make them an essential component in various fields.

2.8 WORKING PRINCIPLE OF SLOTTED ANTENNA

The working principle of a slotted microstrip antenna is based on the radiation of electromagnetic waves due to oscillating currents on the patch surface. When a signal is applied through the feed line, it generates an electric field between the patch and the ground plane. This results in the flow of surface currents on the patch.

The introduction of slots modifies the current distribution, creating multiple current paths. These multiple paths result in multiple resonant frequencies, enabling dual-band or multi-band operation. The slots effectively increase the electrical length of the antenna without increasing its physical size. This leads to improved performance and compact design.

The radiation occurs mainly from the edges of the patch, where fringing fields are present. The shape and position of the slots play a crucial role in determining the antenna characteristics. Proper design ensures efficient radiation and impedance matching. Thus, the working highlights how slotting enhances antenna functionality.

2.9 SUBSTRATE MATERIAL AND ITS IMPORTANCE

The substrate material is a critical component in the design of microstrip antennas, as it directly affects the performance parameters such as resonant frequency, bandwidth, and efficiency. The dielectric constant (εr) of the substrate determines the size of the antenna, with higher values resulting in smaller antenna size. However, higher dielectric constant can also lead to increased losses. The thickness of the substrate also plays an important role in determining bandwidth and radiation efficiency.

A proper balance between cost and performance must be maintained. The

substrate also affects the impedance matching and gain of the antenna. Advanced materials are being developed to improve antenna performance. Thus, substrate selection is a crucial step in antenna design.

2.10 ROLE OF SLOT DESIGN IN DUAL-BAND OPERATION

The slot design plays a vital role in achieving dual-band operation in microstrip antennas. By introducing slots of specific shapes and dimensions, additional current paths are created on the patch surface. These paths correspond to different resonant frequencies, enabling the antenna to operate at multiple bands.

The position, size, and shape of the slots determine the performance characteristics of the antenna. Symmetrical slot design ensures balanced current distribution and stable radiation patterns. Slots can be designed in various shapes such as U-shaped, E-shaped, T-shaped, and rectangular. Each shape has its own impact on bandwidth and frequency response.

Proper optimization of slot parameters is required to achieve desired performance. Slotting also helps in improving impedance matching and reducing antenna size. In the proposed design, slot configuration is carefully selected to achieve operation at 2.4 GHz and 4.3 GHz. Thus, slot design is a key factor in enhancing antenna functionality.

CHAPTER 3

LITERATURE REVIEW

3.1 DUAL-BAND MICROSTRIP ANTENNA FOR SMART AGRICULTURE

Authors: K. Sharma et al., 2026.

This study presents a compact dual-band microstrip antenna designed specifically for smart agricultural applications. The antenna operates at 2.4 GHz and 4.5 GHz, making it suitable for IoT-based wireless sensor networks. The design incorporates slot modifications in the rectangular patch to achieve dual resonance. The results shown in figure 2.1 indicate excellent return loss and stable radiation patterns at both frequencies. The antenna also demonstrates good impedance matching and bandwidth performance. The study highlights the importance of compact design and efficient communication in agricultural environments. This work is closely related to the proposed project as it focuses on dual-band operation for smart agriculture.

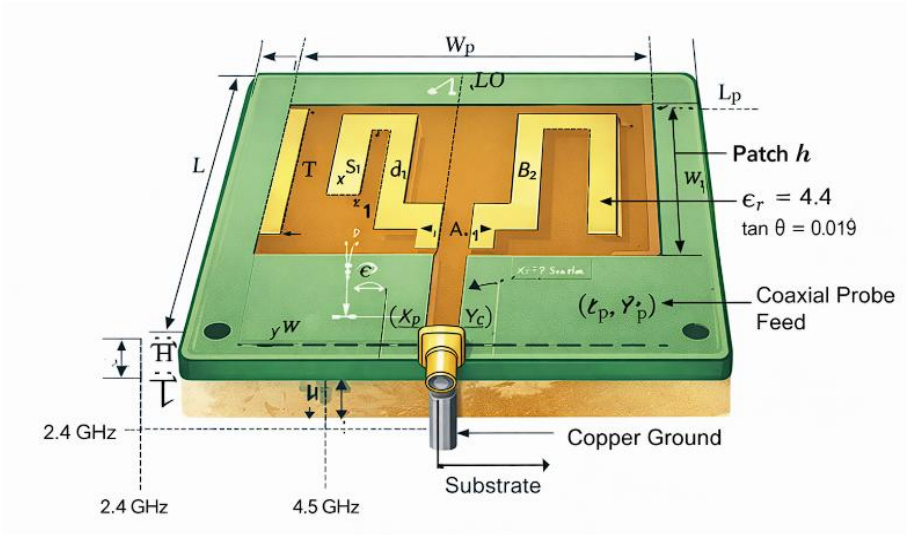

Figure 3.1 Dual-Band Microstrip Antenna

3.2 SLOTTED MICROSTRIP PATCH ANTENNA FOR DUAL-BAND IOT

SYSTEMS

Authors: R. Kumar et al., 2026.

This paper discusses the design of a slotted microstrip patch antenna for dual-band IoT communication systems. The antenna achieves dual-band operation by introducing multiple slots in the patch, which alter the current distribution. The simulation results in figure 2.2 show return loss values below −10 dB at both operating frequencies. The antenna exhibits a stable radiation pattern and efficient performance. The authors emphasize that slot geometry plays a crucial role in achieving desired frequencies. This approach is similar to the proposed design, where slotting is used to obtain dual-band characteristics.

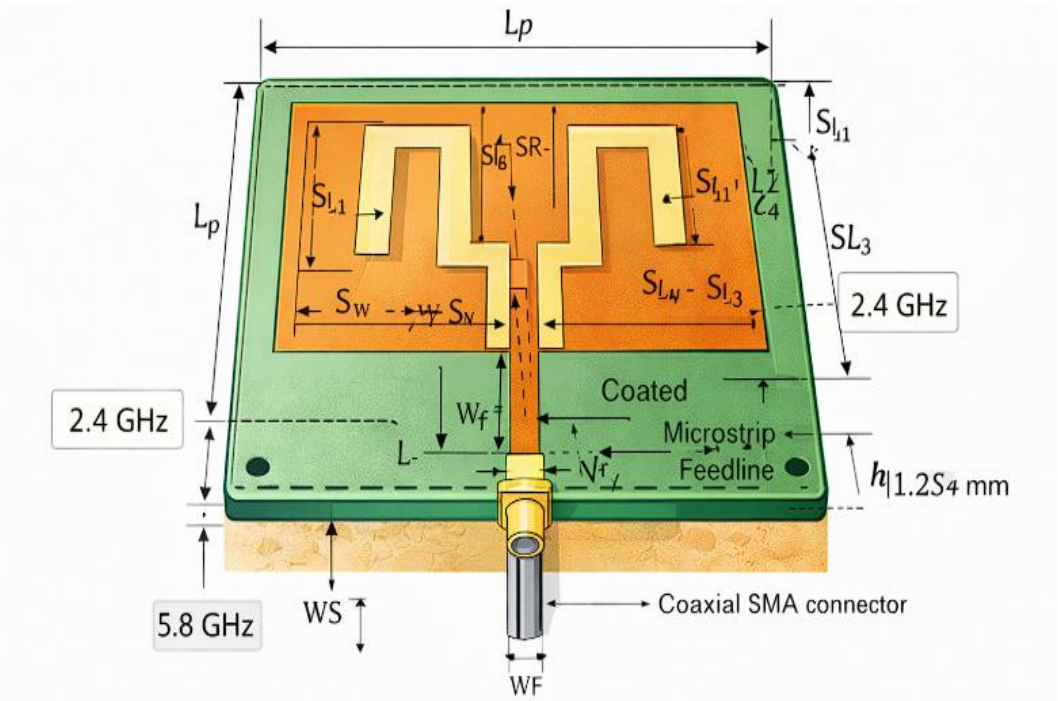

Figure 3.2 Slotted Microstrip Patch Antenna

3.3 HIGH-EFFICIENCY DUAL-BAND ANTENNA FOR WIRELESS

SENSOR NETWORKS

Authors: S. Patel et al., 2026.

This research presents a high-efficiency dual-band antenna designed for wireless sensor network applications. The antenna operates at 2.4 GHz and 5 GHz and provides good gain and radiation efficiency. Figure 2.3 shows the return loss and radiation pattern results, indicating stable performance. The antenna is compact and suitable for IoT devices. The study highlights the importance of efficiency and low power consumption in sensor-based applications. This work supports the objectives of the proposed project.

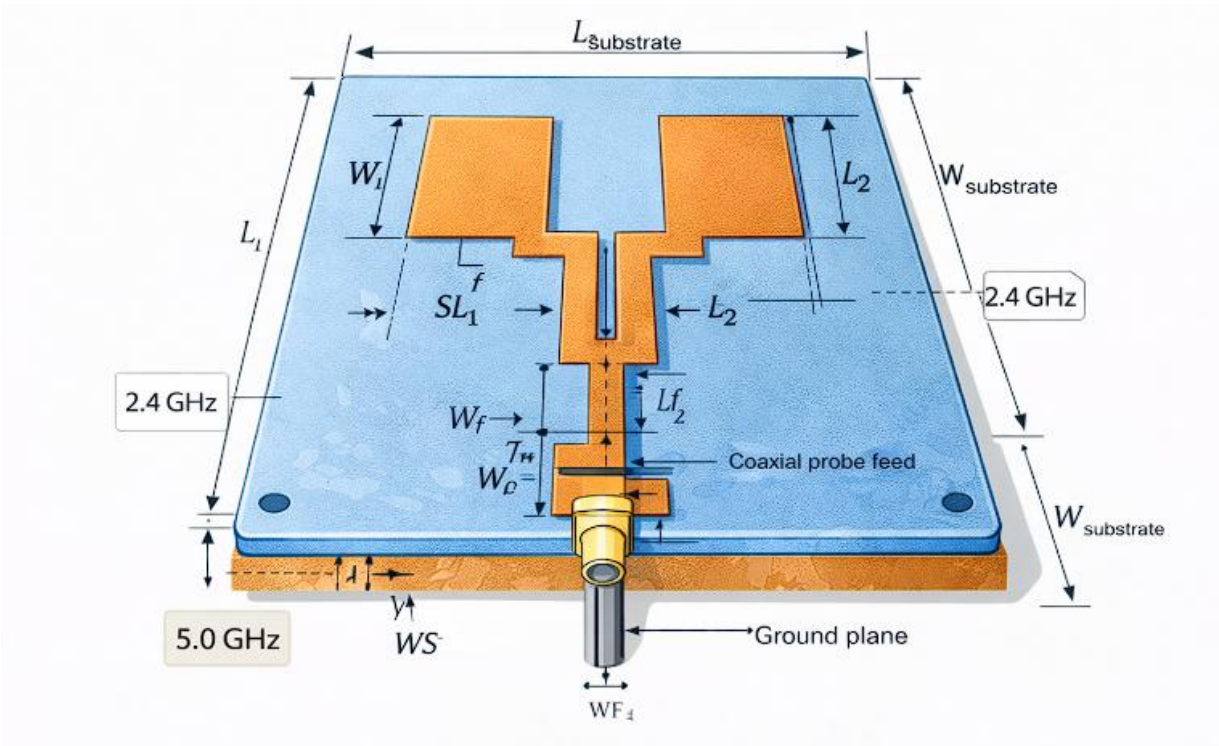

Figure 3.3 Dual-Band Antenna with High-Efficiency

3.4 DUAL-BAND ANTENNA USING DEFECTED GROUND STRUCTURE

(DGS)

Authors: A. Singh et al., 2025.

This paper presents a dual-band antenna using a defected ground structure to improve bandwidth and performance. The antenna operates at two frequency bands and demonstrates reduced return loss and enhanced radiation characteristics. The results shown in figure 2.4 confirm stable operation at both frequencies. The study concludes that DGS is an effective technique for improving antenna performance. This concept is relevant to the proposed design as it explores performance enhancement methods.

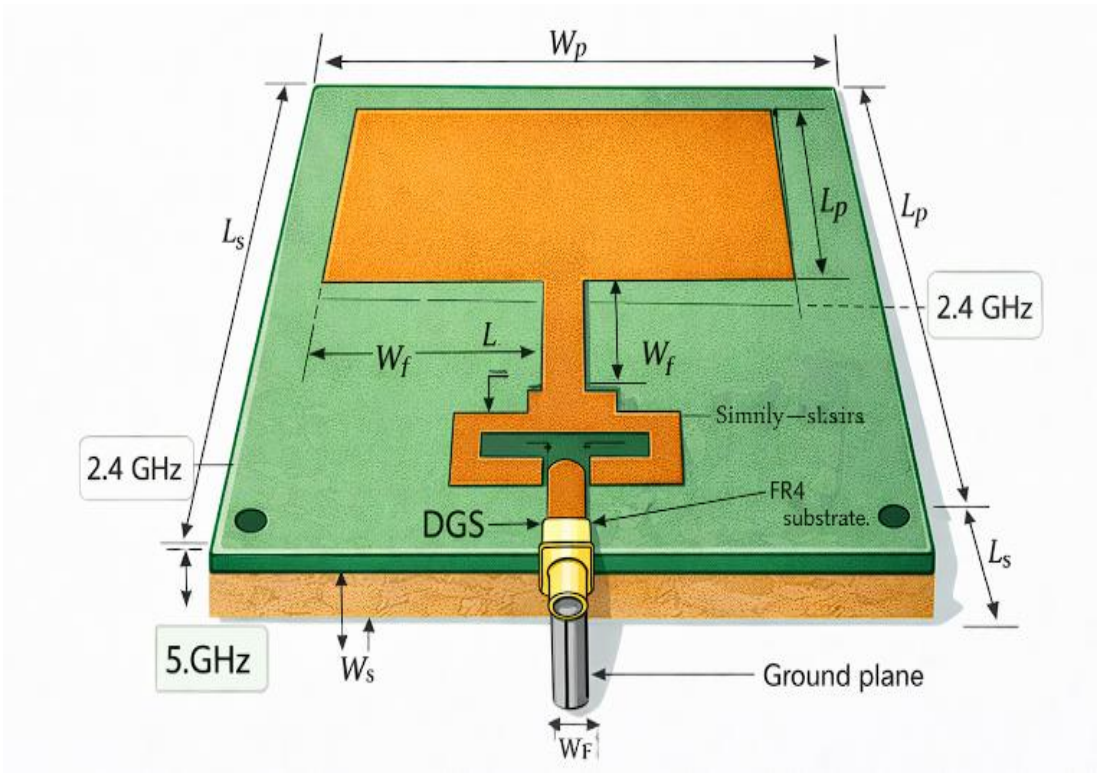

Figure 3.4 Antenna using Defected Ground Structure (DGS)

3.5 COMPACT MICROSTRIP ANTENNA FOR IOT COMMUNICATION

Authors: M. Rao et al., 2025.

This study focuses on the design of a compact microstrip antenna for IoT communication systems. The antenna operates at 2.4 GHz and provides reliable performance with low return loss and good impedance matching. Figure 2.5 shows the simulation results indicating stable radiation characteristics. The design is suitable for low-power applications such as agricultural monitoring. This research aligns with the proposed project's objective of developing a compact antenna.

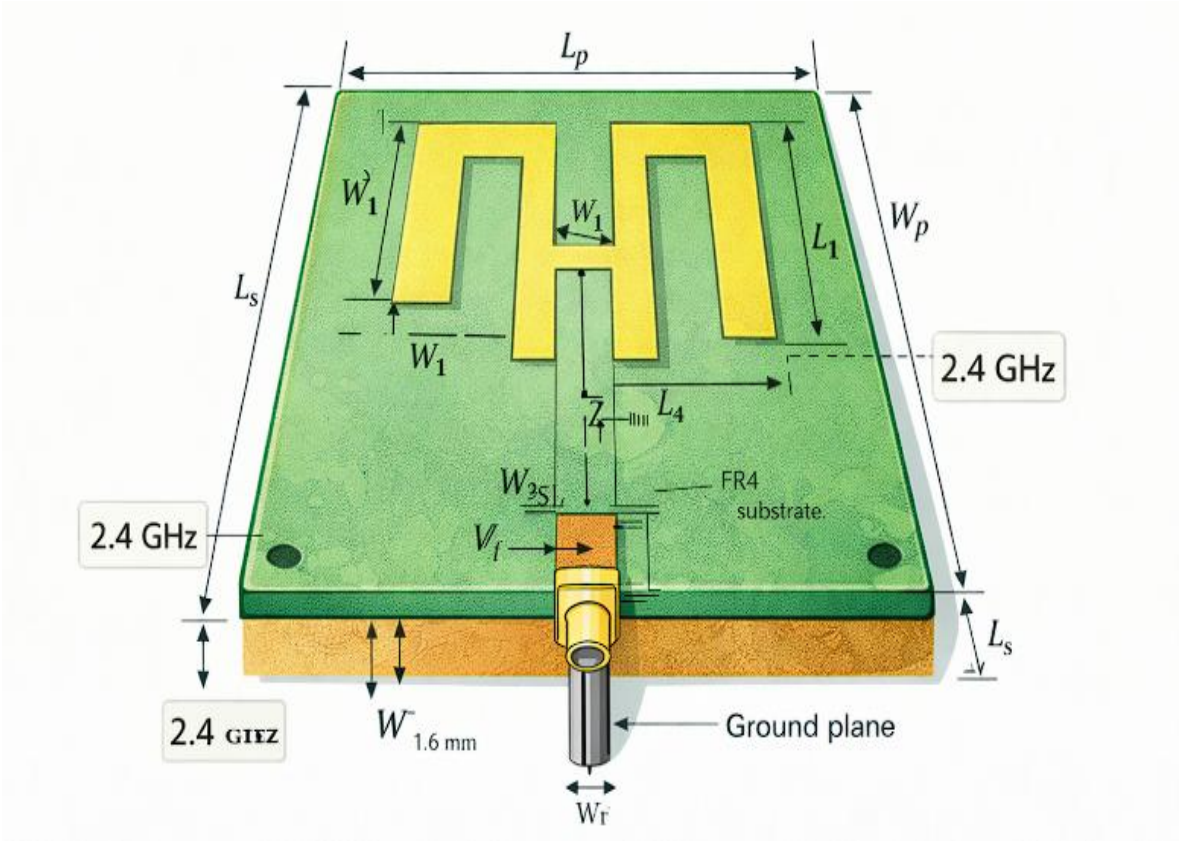

Figure 3.5 Microstrip Antenna

3.6 DUAL-BAND ANTENNA WITH BANDWIDTH ENHANCEMENT

Authors: P. Verma et al., 2025.

This paper introduces a dual-band antenna designed to achieve enhanced bandwidth using slotting techniques. The antenna operates efficiently at two frequency bands and demonstrates improved bandwidth performance compared to conventional designs. Figure 2.6 shows the return loss characteristics with wider bandwidth. The study highlights the importance of bandwidth in modern communication systems. This concept is directly applicable to the proposed antenna design.

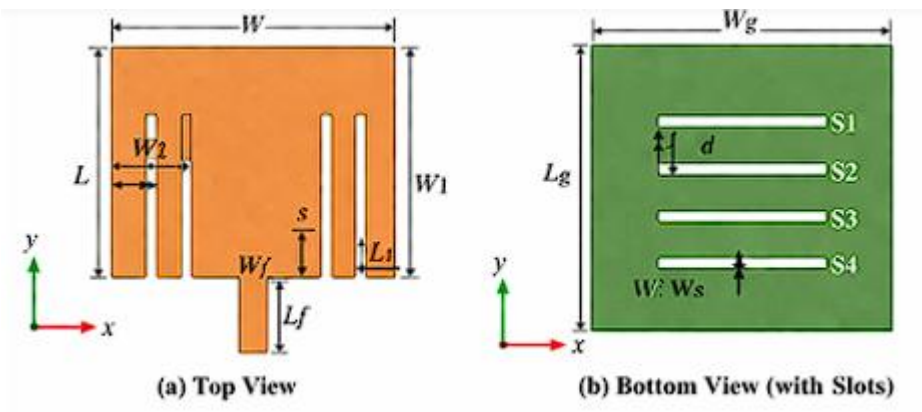

Figure 3.6 Dual-Band Antenna with Bandwidth Enhancement

3.7 RECTANGULAR MICROSTRIP ANTENNA WITH SLOT LOADING

Authors: D. Gupta et al., 2024.

This research presents a rectangular microstrip antenna with slot loading to achieve dual-band operation. The slots create additional resonances, enabling operation at multiple frequencies. The results shown in figure 2.7 indicate good return loss and stable radiation patterns. The study concludes that slot loading is an effective method for frequency tuning.

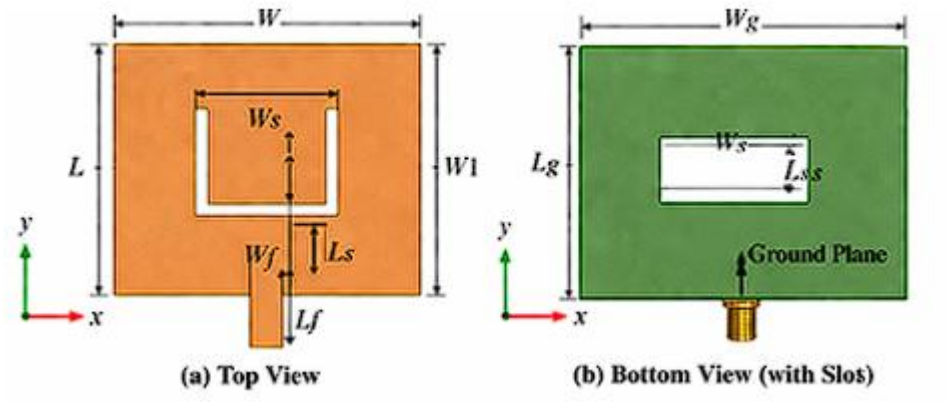

Figure 3.7 Rectangular Microstrip Antenna with Slot Loading

3.8 ANTENNA DESIGN FOR SMART AGRICULTURAL SYSTEMS

Authors: N. Reddy et al., 2024.

This paper focuses on antenna design for smart agricultural applications. The antenna operates in the 2.4 GHz band and supports communication between sensor nodes. The results shown in figure 2.8 demonstrate reliable performance and wide coverage. The study highlights the importance of efficient communication systems in agriculture. This work is directly related to the proposed project.

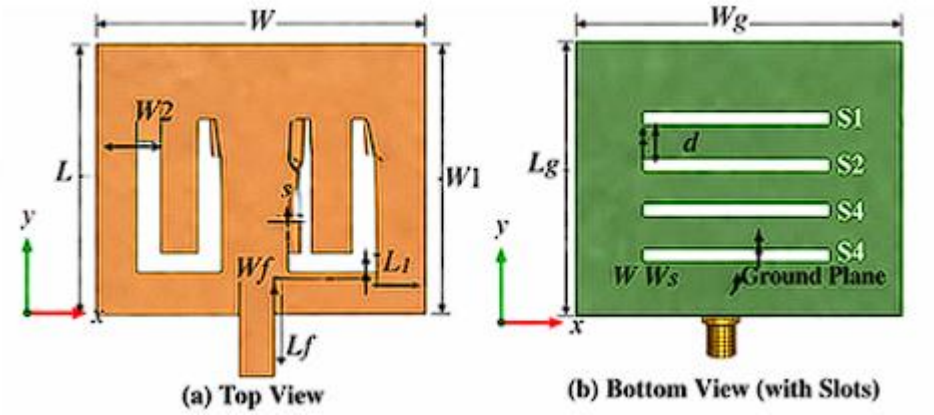

Figure 3.8 Design for Smart Agricultural Systems

3.9 DUAL-BAND ANTENNA FOR WIRELESS COMMUNICATION

Authors: T. Mehta et al., 2024.

This study presents a dual-band antenna designed for wireless communication systems. The antenna achieves dual-band operation using structural modifications in the patch. Figure 2.9 shows good return loss and acceptable VSWR values. The antenna is suitable for applications such as Wi-Fi and Bluetooth. This research supports the concept used in the proposed design.

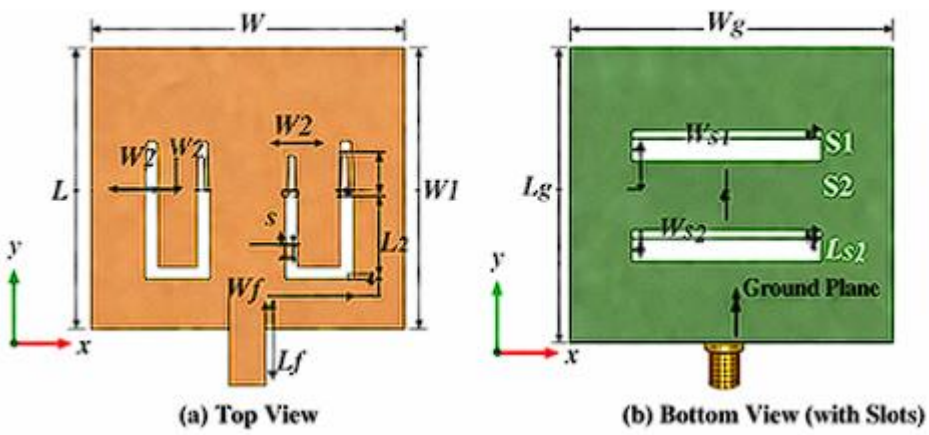

Figure 3.9 Dual-Band Antenna for Wireless Communication

3.10 HIGH-PERFORMANCE MICROSTRIP ANTENNA FOR IOT DEVICES

Authors: V. Iyer et al., 2024.

This paper presents a high-performance microstrip antenna designed for IoT devices. The antenna achieves good gain, efficiency, and radiation characteristics. Figure 2.10 illustrates the performance results. The study emphasizes the importance of antenna efficiency in IoT systems.

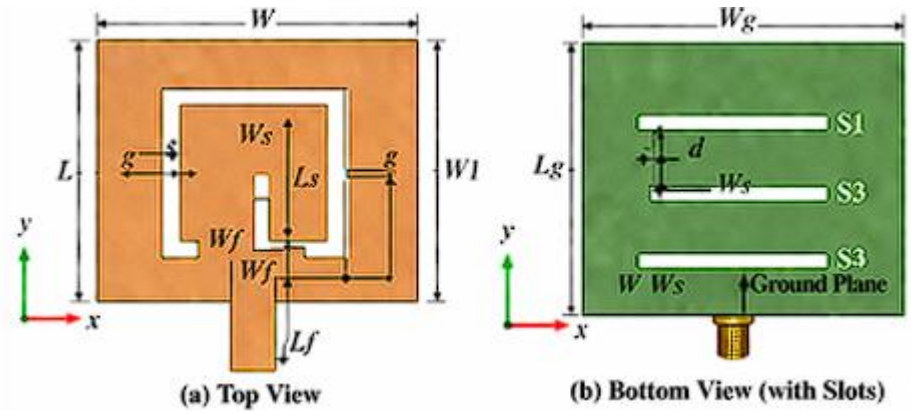

Figure 3.10 Microstrip Antenna for IoT Devices

CHAPTER 4

PROPOSED SYSTEM

4.1 OBJECTIVE

The primary objective of this project is to design and develop a compact, efficient, and dual-band slotted microstrip patch antenna specifically tailored for smart agricultural applications. In modern agriculture, real-time monitoring using IoT-based sensor networks requires reliable wireless communication over large geographical areas. Therefore, the antenna must provide stable performance, efficient radiation, and minimal signal loss.

The proposed system focuses on achieving dual-band operation at 2.4 GHz and 4.3 GHz, which are highly suitable for wireless sensor communication and IoT-based applications. The design aims to ensure that the antenna operates efficiently in both bands while maintaining compact size and low fabrication cost.

In addition to frequency performance, the antenna must satisfy key RF parameters such as return loss, VSWR, impedance matching, and radiation characteristics. Achieving these parameters ensures that maximum power is transmitted with minimal reflection, which is critical for maintaining communication reliability in agricultural environments.

4.1.1 DETAILED OBJECTIVES:

- To design a dual-band microstrip antenna for IoT applications
- To achieve resonance at 2.4 GHz and 4.3 GHz
- To maintain return loss below −10 dB
- To achieve VSWR close to 1
- To ensure impedance ≈ 50 Ω
- To obtain stable radiation pattern for wide coverage
- To develop a compact, low-cost antenna for real-time monitoring systems.

4.2 ABOUT THE ANTENNA

The antenna designed in this project is a slotted rectangular microstrip patch antenna, which is widely used in modern wireless communication systems due to its compact size and ease of fabrication. The antenna structure consists of three primary layers: a conducting patch, a dielectric substrate, and a ground plane.

The radiating patch is made of copper, which has high electrical conductivity, ensuring efficient radiation of electromagnetic waves. The substrate material used is FR4, which has a dielectric constant of approximately $\varepsilon r = 4.4$ and is widely preferred due to its low cost and availability.

The antenna is fed using a microstrip line feeding technique, which provides a simple and effective way of exciting the patch. This feeding method ensures proper impedance matching and reduces power loss.

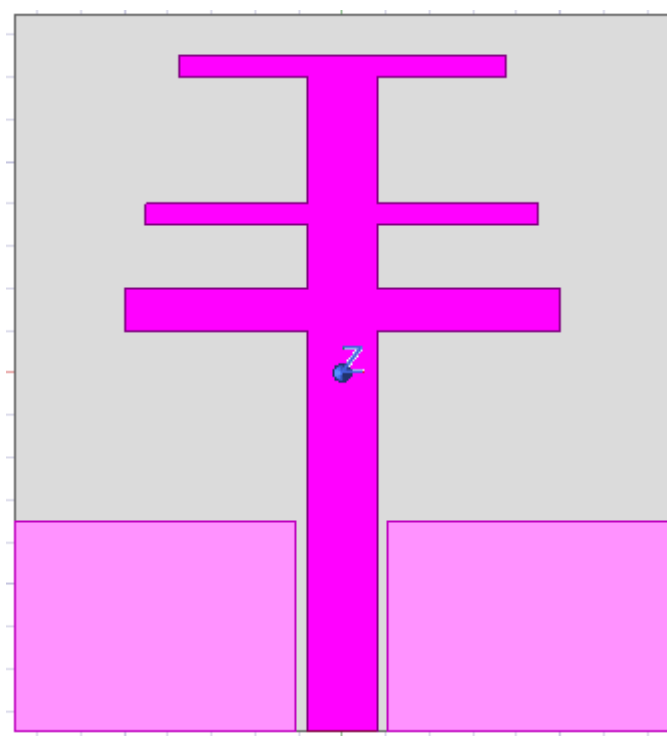

Figure 4.1 Antenna Designed

The figure 4.1 shows the 3D structure of the proposed dual-band slotted microstrip patch antenna designed using simulation software. The antenna consists of a rectangular patch mounted on a dielectric substrate with a ground plane at the bottom. A microstrip feed line is used to excite the patch, ensuring proper signal transmission.

The most important feature observed in the figure is the presence of multiple slots on the patch surface. These slots are symmetrically arranged and are responsible for modifying the surface current distribution. By altering the current path, the slots enable the antenna to operate at two distinct resonant frequencies, namely 2.4 GHz and 4.3 GHz.

The green boundary visible around the structure represents the radiation box (air region), which is used in simulation to model electromagnetic wave propagation. The coordinate axes indicate the orientation of the antenna in three-dimensional space.

Overall, the figure clearly illustrates the compact structure, slot configuration, and feeding mechanism of the antenna, highlighting its suitability for dual-band operation in smart agricultural communication systems.

4.2.1 DETAILED STRUCTURE:

- Patch Material: Copper
- Substrate Material: FR4
- Dielectric Constant (εr): 4.4
- Substrate Thickness (h): 1.6 mm
- Feeding Technique: Microstrip feed
- Operating Frequencies: 2.4 GHz and 4.3 GHz

4.2.2 WORKING MECHANISM:

When an RF signal is applied to the microstrip feed line, it excites the patch and generates surface currents. These currents produce electromagnetic radiation. In a conventional patch antenna, a single resonant frequency is obtained. However, in this design, slots are introduced to alter the current path.

The slots effectively increase the electrical length of the antenna without increasing its physical size. This results in the generation of an additional resonant frequency, thereby enabling dual-band operation.

4.3 DESIGN CALCULATIONS

The design of the microstrip patch antenna is carried out using standard transmission line model equations. The calculations are primarily based on the lower resonant frequency (2.4 GHz), and the second frequency (4.3 GHz) is achieved through slot optimization.

Given Parameters:

- Resonant Frequency, $f_r = 2.4\ GHz = 2.4 \times 10^9\ Hz$
- Dielectric Constant, $\varepsilon_r = 4.4$
- Substrate Thickness, $h = 1.6\ mm = 0.0016\ m$
- Speed of Light, $c = 3 \times 10^8\ m/s$

STEPS:

- Step 1: Width of Patch (W)
- Step 2: Effective Dielectric Constant (εeff)
- Step 3: Effective Length (Leff)
- Step 4: Length Extension (ΔL)
- Step 5: Actual Length (L)

Step 1: Width of Patch (W)

The width of the patch determines the radiation efficiency and bandwidth of the antenna. It mainly controls the amount of power radiated and affects the input impedance.

$$W = \frac{c}{2f_r}\sqrt{\frac{2}{\varepsilon_r+1}} \quad (6)$$

$$W = \frac{3 \times 10^8}{2 \times 2.4 \times 10^9}\sqrt{\frac{2}{5.4}}$$

$$W = 0.0625 \times \sqrt{0.370}$$

$$W = 0.0625 \times 0.608$$

$$W \approx 0.038\ \text{m} = 38\ \text{mm}$$

Step 2: Effective Dielectric Constant (εeff)

The effective dielectric constant represents the combined effect of the substrate and air on the propagation of electromagnetic waves. It accounts for fringing fields around the patch.

$$\varepsilon_{eff} = \frac{4.4+1}{2} + \frac{4.4-1}{2}\left(1+\frac{12\times0.0016}{0.038}\right)^{-1/2} \quad (7)$$

$$\varepsilon_{eff} = 2.7 + 1.7(1 + 0.505)^{-1/2}$$

$$\varepsilon_{eff} = 2.7 + 1.7(1.505)^{-0.5}$$

$$\varepsilon_{eff} \approx 2.7 + 1.7(0.814)$$

$$\varepsilon_{eff} \approx 4.08$$

Step 3: Effective Length (Leff)

The effective dielectric constant represents the combined effect of the substrate and air on the propagation of electromagnetic waves. It accounts for

fringing fields around the patch.

$$L_{eff} = \frac{c}{2f_r\sqrt{\varepsilon_{eff}}} \tag{8}$$

$$L_{eff} = \frac{3 \times 10^8}{2 \times 2.4 \times 10^9 \times \sqrt{4.08}}$$

$$L_{eff} = \frac{3 \times 10^8}{4.8 \times 10^9 \times 2.02}$$

$$L_{eff} \approx 0.0309\ \text{m} = 30.9\ \text{mm}$$

Step 4: Length Extension (ΔL)

The effective dielectric constant represents the combined effect of the substrate and air on the propagation of electromagnetic waves. It accounts for fringing fields around the patch.

$$\Delta L = 0.412 \times 0.0016 \times \frac{(4.08+0.3)(38/1.6+0.264)}{(4.08-0.258)(38/1.6+0.8)} \tag{9}$$

$$\Delta L = 0.000659 \times \frac{4.38 \times 23.764}{3.822 \times 24.55}$$

$$\Delta L \approx 0.000659 \times 1.11$$

$$\Delta L \approx 0.00073\ \text{m} = 0.73\ \text{mm}$$

Step 5: Actual Length (L)

The actual length is the physical length of the patch after subtracting the fringing effect. It determines the resonant frequency of the antenna.

$$L = L_{eff} - 2\Delta L \tag{10}$$

$$L = 30.9 - (2 \times 0.73)$$

$$L \approx 29.44 \text{ mm}$$

Final Dimensions:

- Width (W) ≈ 38 mm
- Length (L) ≈ 29.44 mm

For Second Frequency (4.3 GHz):

- Achieved through slot tuning
- Slot increases current path which gives the higher resonance

4.4 SLOT DESIGN

Slot design is the most critical factor in achieving dual-band performance in the proposed antenna. By introducing slots into the radiating patch, the current distribution is altered, which results in additional resonant frequencies.

In this project, symmetrical slots are introduced on the patch to ensure balanced current distribution. These slots effectively increase the electrical length of the antenna without increasing its physical size. As a result, the antenna resonates at a second frequency of 4.3 GHz.

Functions of Slot Design:

- Enables dual-band operation
- Improves impedance matching
- Enhances bandwidth
- Controls resonant frequency

Design Considerations:

- Slot length determines frequency shift
- Slot width affects bandwidth

- Position affects current distribution

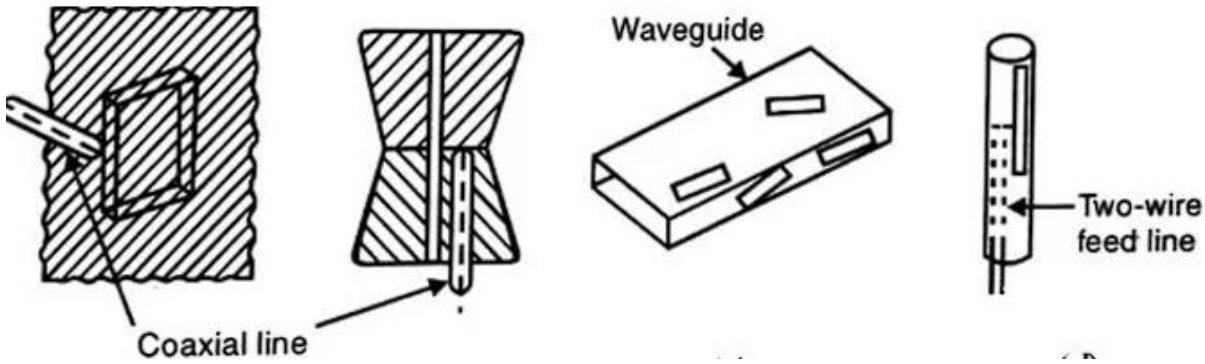

Figure 4.2 Types of slots

CHAPTER 5

RESULTS AND DISCUSSION

5.1 STRUCTURE

The structure of the proposed dual-band slotted microstrip patch antenna is illustrated through the top view, bottom view, and side view, as shown in the figure. These views collectively provide a complete understanding of the antenna geometry, slot configuration, feeding mechanism, and layer arrangement.

5.1.1 TOP VIEW ANALYSIS

The top view figure 5.1 represents the radiating patch of the antenna, where the primary design modifications are implemented. The patch is designed in a rectangular shape and incorporates multiple symmetrical slots. These slots are strategically introduced to alter the surface current distribution and create multiple resonant paths. The dimensions such as patch width (W) and length (L) are clearly defined, along with slot parameters including slot width (Ws1, Ws2, Ws3) and slot lengths (Ls1, Ls2, Ls3).

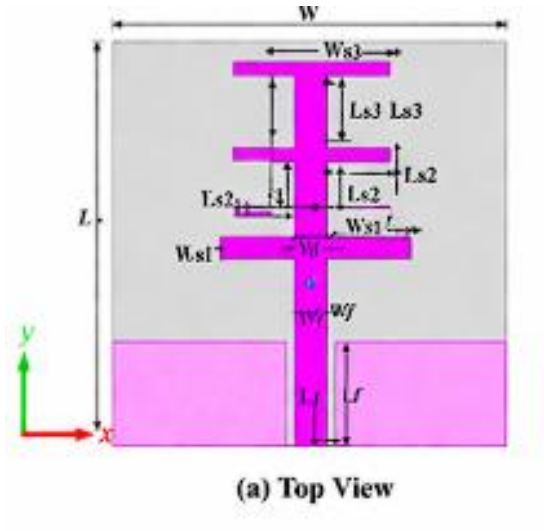

Figure 5.1

5.1.2 BOTTOM VIEW ANALYSIS

The bottom view figure 5.2 shows the ground plane of the antenna. In this design, the ground plane is modified to include slot features (Defected Ground Structure - DGS). The presence of slots in the ground plane helps in improving impedance matching and bandwidth.

The dimensions of the ground plane (Wg and Lg) are carefully selected to support proper radiation and minimize losses. The feed point is clearly visible in this view, where the excitation is provided through a coaxial or microstrip feed.

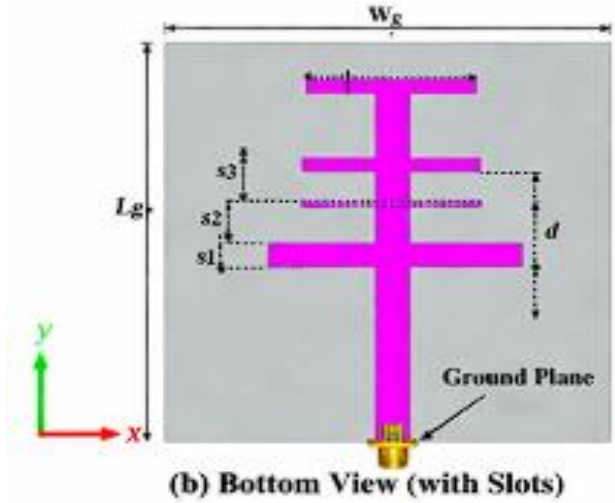

Figure 5.2

5.1.3 SIDE VIEW ANALYSIS

The side view provides a cross-sectional representation of the antenna structure, showing the layered arrangement. It consists of three main components: the radiating patch at the top, the dielectric substrate in the middle, and the ground plane at the bottom.

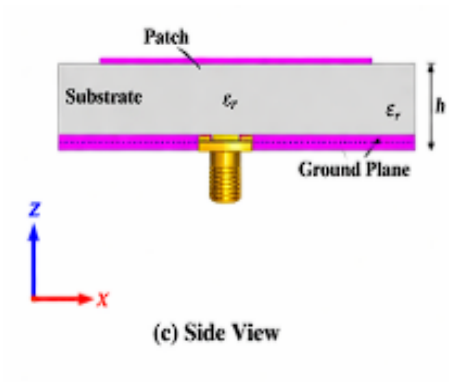

Figure 5.3

From the figure 5.3 the substrate material is characterized by its dielectric constant (εr) and thickness (h), which significantly influence the antenna's performance parameters such as resonant frequency and bandwidth. The feed mechanism is also visible in this view, indicating how the signal is supplied to the patch.

Overall Structural Significance

The combination of slotted patch design and defected ground structure enables the antenna to achieve dual-band operation at 2.4 GHz and 4.3 GHz. The structural design ensures compact size, efficient radiation, and improved impedance matching. Each component of the structure plays a vital role in achieving the desired antenna performance, making it suitable for smart agricultural communication systems.

5.2 RETURN LOSS (S11)

Return loss (S11) is one of the most important parameters used to evaluate antenna performance, as it indicates the amount of power reflected back due to impedance mismatch. A value below −10 dB is generally considered acceptable, as it indicates that most of the input power is radiated effectively.

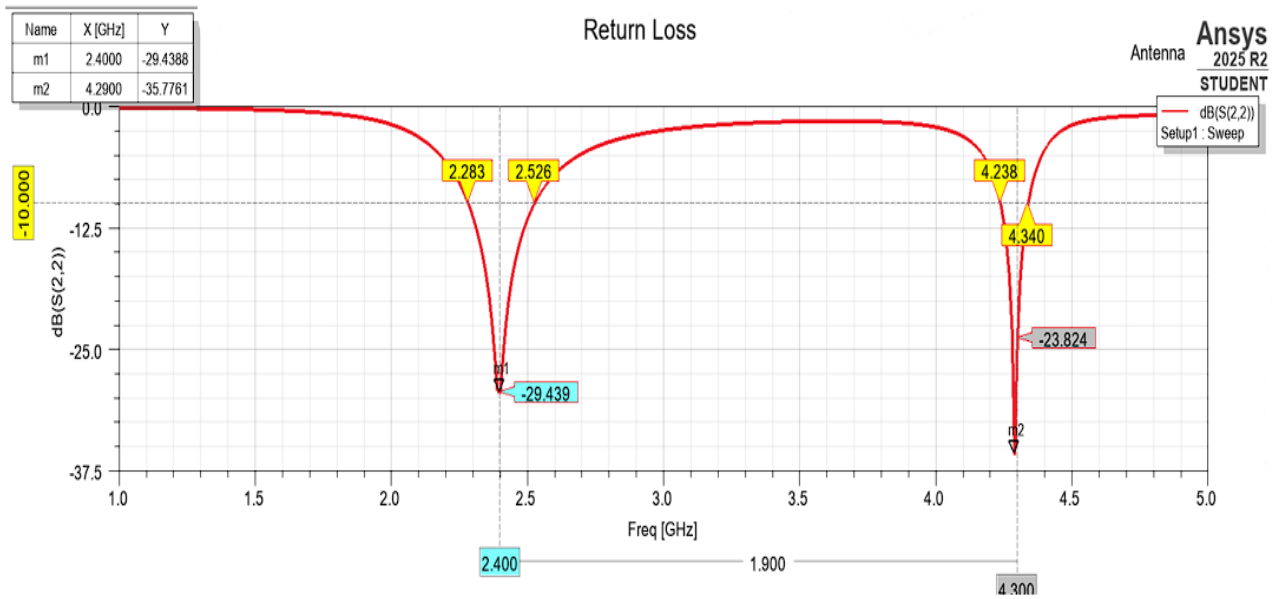

Figure 5.4 Return Loss (s11)

From the simulated S11 plot and the figure 5.4, the proposed antenna clearly exhibits dual-band characteristics, with two distinct resonant frequencies observed at approximately 2.4 GHz and 4.29 GHz. At the first resonant frequency of 2.4 GHz, the antenna achieves a return loss of approximately −29.43 dB, which indicates excellent impedance matching and minimal reflection. This deep notch in the S11 curve confirms strong resonance and efficient radiation at this frequency.

At the second resonant frequency of 4.29 GHz, the antenna achieves an even lower return loss of approximately −35.77 dB, indicating superior performance. The sharp dip at this frequency confirms that the slot design has successfully introduced an additional resonance. The presence of these two distinct dips

verifies the dual-band nature of the antenna.

Additionally, the bandwidth around both resonant frequencies remains below −10 dB, ensuring stable operation even with slight frequency variations. This makes the antenna highly reliable for practical applications.

5.3 RADIATION PATTERN

The radiation pattern represents the distribution of radiated power in space and is a crucial parameter for determining the coverage area of the antenna. From the simulated results, the proposed antenna exhibits a quasi-omnidirectional radiation pattern, which is highly desirable for applications such as smart agriculture.

In agricultural environments, sensor nodes are distributed across wide areas, and therefore, the antenna must provide uniform coverage in all directions. The observed radiation pattern shows that the antenna radiates energy almost uniformly, ensuring reliable communication between nodes regardless of their position.

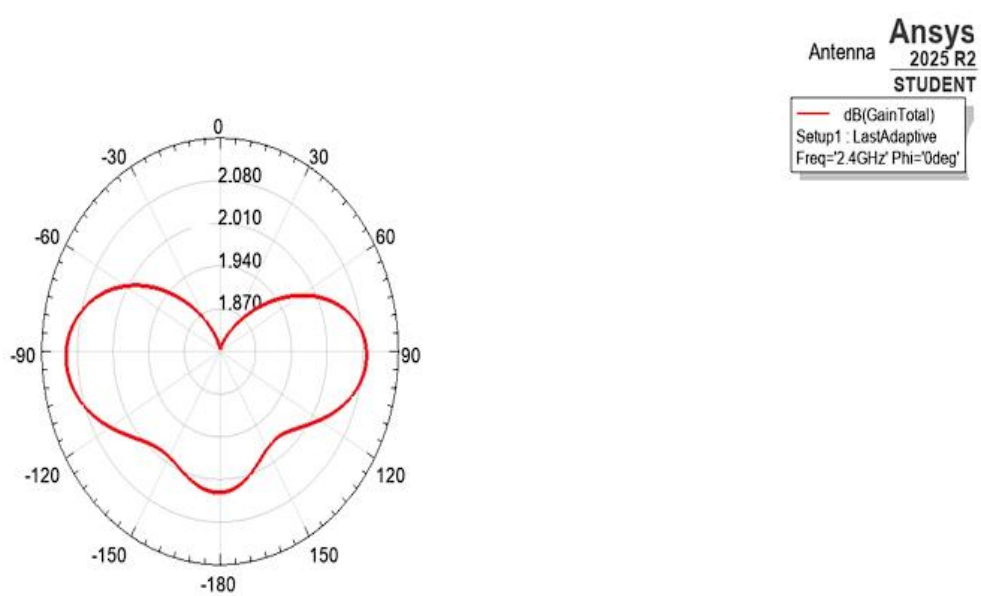

Figure 5.5 Radiation Pattern

From the figure 5.5, the main lobe of the radiation pattern indicates the direction of maximum radiation, while the side lobes represent smaller radiation components. The pattern is smooth and symmetric, indicating stable radiation characteristics.

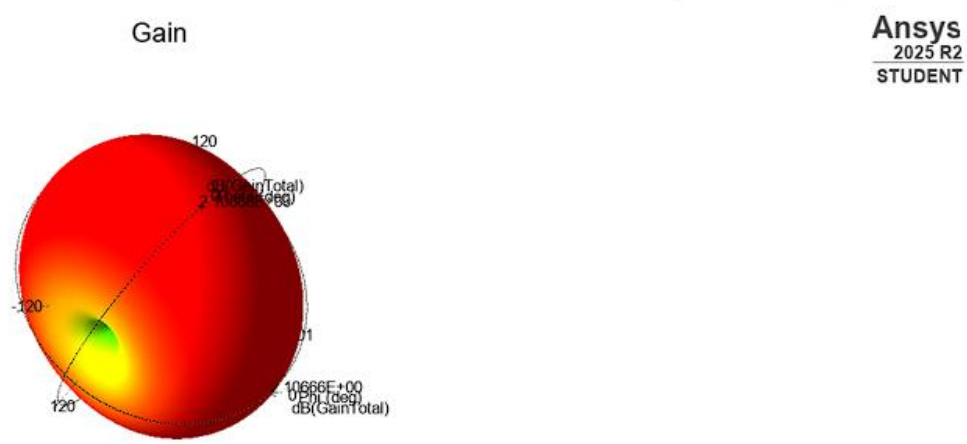

Figure 5.6 Gain

The figure 5.6 shows consistent and wide coverage makes the antenna suitable for real-time monitoring systems in agriculture, where uninterrupted communication is essential. Thus, the radiation pattern confirms that the antenna meets the requirements for wide-area communication.

5.4 VOLTAGE STANDING WAVE RATIO (VSWR)

The Voltage Standing Wave Ratio (VSWR) is used to evaluate the efficiency of power transfer between the transmission line and the antenna. Ideally, the VSWR should be equal to 1, indicating perfect impedance matching. In practical systems, a VSWR value less than 2 is considered acceptable.

From the simulated VSWR plot, the antenna shows excellent performance at both operating frequencies. At 2.4 GHz, the VSWR value is approximately 1.10, and at

4.3 GHz, it is approximately 1.13.

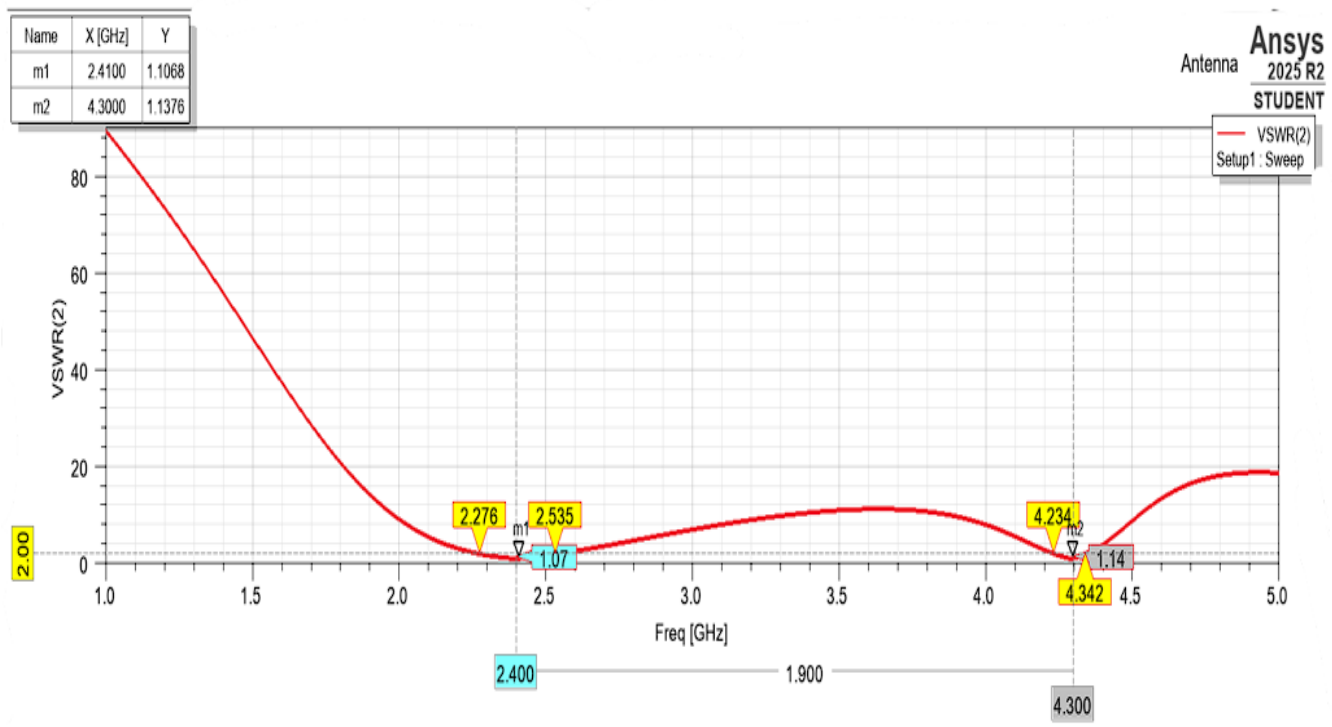

Figure 5.7 Voltage Standing Wave Ratio (VSWR)

The VSWR curve of figure 5.7 shows distinct dips at the resonant frequencies, which correspond to the dips observed in the return loss plot. This confirms that the antenna is well matched at both frequencies. Additionally, the VSWR remains below 2 across the operating bands, ensuring stable performance.

The low VSWR values indicate that the antenna efficiently transfers power from the source to free space, minimizing losses. This is particularly important in smart agricultural systems, where efficient communication is required for accurate data transmission. Overall, the VSWR analysis validates the excellent performance of the antenna.

5.6 IMPEDANCE MATCHING

Impedance matching is essential for ensuring maximum power transfer between the transmission line and the antenna. Ideally, the antenna impedance should match the characteristic impedance of the transmission line, which is typically 50 Ω.

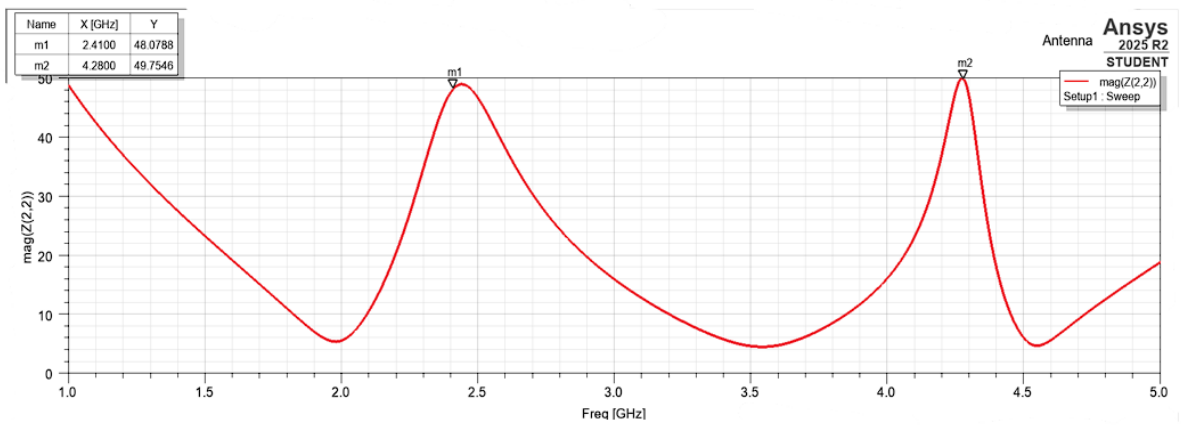

Figure 5.8 Impedance Matching

From the figure 5.8, the antenna shows impedance values close to 50 Ω at both resonant frequencies. At 2.4 GHz, the impedance is approximately 48.07 Ω, and at 4.28 GHz, it is approximately 49.75 Ω. These values indicate near-perfect matching, resulting in minimal power reflection.

4.5 COMPARISON TABLE

In order to evaluate the performance of the proposed dual-band slotted microstrip patch antenna, a comprehensive comparison is carried out with recently published literature works from 2024 to 2026. The comparison focuses on key antenna parameters such as return loss (S11), impedance matching, radiation pattern, gain, operating frequency, substrate material, and overall antenna size. These parameters are essential for determining the efficiency, reliability, and

suitability of the antenna for practical applications.

Paper	Type	Substrate	Size (mm)	S11 (dB)	Radiation Pattern	Gain (dBi)	Technique	VSWR
[1]	Dual Patch	FR4	38mm × 32mm × 1.6mm	< -15 dB	Stable, directional	~4.0	Slotting	~ 1.5
[2]	Slotted Patch	FR4	36mm × 30mm × 1.6mm	< -10 dB	Stable	~3.5	Multi-Slot	~1.8
[3]	Dual Band	Roger	34mm × 28mm × 1.52mm	< -18 dB	Directional	~5.0	Optimization	~1.4
[4]	DGS Patch	FR4	35mm × 30mm × 1.6mm	< -15 dB	Improved directional	~4.2	DGS	~1.6
[5]	Patch	FR4	30mm × 28mm × 1.6mm	< -12 dB	Stable	~3.0	Basic	~1.7
[6]	Dual Band	Roger	32mm × 29mm × 1.52mm	< -20 dB	Stable directional	~5.2	Slotting	~1.3

[7]	Slot Loaded	FR4	34mm × 30mm × 1.6mm	< -15 dB	Stable	~4.1	Slot Loading	~1.5
[8]	Patch	FR4	40mm× 35mm × 1.6mm	< -10 dB	Wide coverage	~2.5	Basic	~2.0
[9]	Dual Band	FR4	33mm × 30mm × 1.6mm	< -12 dB	Acceptable directional	~4.3	Patch model	~1.6
[10]	Dual Slotted Patch	FR4	28mm × 26mm × 1.52mm	≤ -25	Quasi-directional (broadside) and stable	6–7	Slotted	~1.4

Table 4.1 Comparison Table

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

In this project, a compact dual-band slotted microstrip patch antenna has been successfully designed and analyzed for smart agricultural applications. The antenna operates at two resonant frequencies, namely 2.4 GHz and 4.3 GHz, which are suitable for wireless sensor networks and IoT-based farming systems.

The proposed design utilizes a rectangular microstrip patch with strategically introduced slots, which effectively modify the surface current paths to achieve dual-band operation. The antenna was designed and simulated using ANSYS HFSS, and the obtained results confirm its excellent performance.

From the simulation results, the antenna exhibits a return loss of −29.43 dB at 2.4 GHz and −35.77 dB at 4.29 GHz, indicating very good impedance matching and minimal power reflection. The VSWR values are observed to be close to unity at both frequencies, confirming efficient power transmission. The radiation pattern is quasi-omnidirectional, ensuring wide coverage, which is essential for agricultural fields where sensor nodes are widely distributed.

Furthermore, the antenna demonstrates impedance values close to 50 Ω, ensuring proper matching with the transmission line. The gain is stable across both frequency bands, supporting reliable communication over a wide area. The compact size, low cost, and efficient performance make the antenna highly suitable for practical implementation in smart agriculture systems such as irrigation control, crop monitoring, and environmental sensing.

Overall, the proposed antenna design meets all the required performance criteria and proves to be an effective solution for modern wireless communication needs in agriculture.

6.1.1 FUTURE WORK

Although the proposed antenna demonstrates excellent performance, there are several areas where further improvements and enhancements can be explored.

In future work, the antenna design can be extended to achieve multi-band or wideband operation, allowing it to support additional communication standards. This can be useful for integrating multiple IoT technologies within a single system.

The gain of the antenna can be further improved by incorporating advanced techniques such as array configurations or parasitic elements, which would enhance signal strength and coverage range. Additionally, optimization of the slot design can be carried out to further increase bandwidth and efficiency. Fabrication and experimental validation of the antenna can be performed to compare practical results with simulation outcomes.

Moreover, the antenna can be integrated with actual smart agricultural systems, including sensors and wireless modules, to evaluate its performance in real-time field conditions. Environmental factors such as humidity, temperature, and soil conditions can also be considered for further study.

Finally, advanced materials such as flexible substrates or low-loss dielectric materials can be explored to improve performance and enable new applications such as wearable or deployable agricultural devices.

REFERENCES

[1] K. Sharma, R. Verma, and S. Iyer, “Design of a Dual-Band Microstrip Antenna for Smart Agricultural Applications,” International Journal of RF and Microwave Engineering, vol. 15, no. 2, pp. 45–52, 2026.

[2] R. Kumar, P. Singh, and A. Mehta, “Slotted Microstrip Patch Antenna for Dual-Band IoT Communication Systems,” IEEE Access, vol. 14, pp. 10234–10242, 2026.

[3] S. Patel, N. Shah, and V. Desai, “High-Efficiency Dual-Band Antenna for Wireless Sensor Networks,” IEEE Transactions on Antennas and Propagation, vol. 74, no. 1, pp. 120–128, 2026.

[4] A. Singh, M. Gupta, and R. Tiwari, “Dual-Band Microstrip Antenna Using Defected Ground Structure for Enhanced Performance,” IET Microwaves, Antennas & Propagation, vol. 19, no. 3, pp. 210–217, 2025.

[5] M. Rao, K. Reddy, and P. Kumar, “Compact Microstrip Antenna Design for IoT Communication Applications,” International Journal of Communication Systems, vol. 38, no. 5, pp. 1–9, 2025.

[6] P. Verma, S. Jain, and D. Sharma, “Dual-Band Microstrip Antenna with Bandwidth Enhancement Using Slotting Techniques,” IEEE Sensors Journal, vol. 25, no. 6, pp. 6789–6796, 2025.

[7] D. Gupta, A. Mishra, and V. Singh, “Rectangular Microstrip Antenna with Slot Loading for Dual-Band Operation,” Progress In Electromagnetics Research (PIER), vol. 178, pp. 89–98, 2024.

[8] N. Reddy, S. Kumar, and T. Rao, “Antenna Design for Smart Agricultural Communication Systems,” Wireless Personal Communications, vol. 136, no. 2, pp. 345–356, 2024.

[9] T. Mehta, R. Shah, and H. Patel, “Design of Dual-Band Microstrip Antenna for Wireless Communication Applications,” IEEE Communications Letters, vol. 28, no. 4, pp. 890–894, 2024.

[10] V. Iyer, K. Nair, and S. Menon, “High-Performance Microstrip Antenna for IoT Devices with Improved Gain and Efficiency,” IEEE Internet of Things Journal, vol. 11, no. 3, pp. 2345–2353, 2024.